

BORDER OPTICS — POORI RESEARCH SAMJHO

PDF 4 of 22 — Foundations

Economics/Causal Inference Zero Se Shuru: 'Scheme Ki Wajah Se' Kaise Prove Karte Hain

Sakshi ke liye, Sakshi ki apni research par — samjhaya gaya ekdum scratch se, taaki H4 (control group DiD) test ka poora logic, aur uski real, kabhi-messy history, crystal clear ho.

Is PDF mein: counterfactual ka impossible-experiment problem, control group kyun zaroori hai, Difference-in-Differences (DiD) ka poora logic, parallel trends assumption, district fixed effects, aur ek real, asli kahani — kaise ek control-group data-bug ne ek 'significant' result diya tha jo fix hone ke baad gayab ho gaya. Sab BORDER OPTICS ke asli numbers, asli code, aur asli Development Log entries se.

Shuru Karte Hain — Ek Aisa Sawaal Jo Kabhi Directly Answer Nahi Ho Sakta

PDF 3 mein humne seekha ki statistics ye batati hai ki ek change 'real' hai ya sirf chance. Ab ek aur, usse bhi gehra sawaal aata hai — jo BORDER OPTICS ke naam mein hi chhupa hai: agar kisi gaon mein NDBI badha bhi, kya ye VVP-I scheme ki wajah se badha, ya wo gaon aise hi, apne aap, waqt ke saath badh jaata — koi bhi scheme ho ya na ho?

Ye sawaal 'causal inference' ka core hai, aur ye economics/policy-evaluation ki poori field ka buniyaadi sawaal hai. Sirf ye dekhna ki 'gaon mein development hua' kaafi nahi — humein ye jaanna hai ki kya development is scheme ki wajah se hua, jo bina scheme ke nahi hota.

EXAMPLE — Dawai Ki Analogy

Socho ek naya bukhari ka tablet market mein aaya hai. Ek patient ne tablet li aur 3 din mein theek ho gaya. Kya iska matlab tablet ne kaam kiya? Shayad — lekin bukhari toh kai baar apne aap bhi 3 din mein theek ho jaata hai, bina kisi dawai ke. Sirf 'patient ko dawai di, aur wo theek hua' dekhne se kabhi pata nahi chalega ki dawai ne kuch kiya ya nahi — jab tak hum ye na jaanenge ki agar usne dawai na li hoti, tab kya hota. Yahi exact problem VVP-I ke saath hai — humein pata hai kya hua (villages mein NDBI/lights measure kiya), lekin humein kabhi seedha pata nahi chal sakta ki 'agar VVP-I na hota, toh kya hota' — kyunki wo scenario kabhi hua hi nahi.

1. Counterfactual — Wo 'Doosri Duniya' Jo Kabhi Dekh Nahi Sakte

Is 'agar aisa na hota toh kya hota' wali imaginary duniya ko economics mein counterfactual kehte hain. Har causal question ka core structure yahi hai: asli, jo hua (jaise 'Kaho mein VVP-I mila, aur Kaho mein itna development hua'), aur counterfactual, jo nahi hua lekin agar halaat alag hote toh hota ('agar Kaho ko VVP-I na milta, tab Kaho mein kitna development hota'). Asli aur counterfactual ka farak hi asli, causal effect hai.

Problem ye hai ki counterfactual kabhi directly observe nahi ho sakta — ek hi gaon, ek hi waqt par, dono scenarios (VVP-I mila aur VVP-I nahi mila) mein simultaneously nahi ho sakta. Isko economics mein 'fundamental problem of causal inference' kaha jaata hai — aur poori causal-inference field ka kaam hai iske around clever, indirect tareeke dhundhna.

EXAMPLE — Time-Travel Wali Analogy

Agar humare paas time machine hoti, hum Kaho ko VVP-I dete, uska development note karte, fir time wapas ghumake Kaho ko VVP-I na dete, aur dekhte dobara kya hota — dono outcomes compare kar sakte, aur exact causal effect mil jaata. Lekin time machine exist nahi karti. Toh humein ek 'proxy' counterfactual chahiye — kuch aisa jo Kaho jaisa hi ho, lekin jisko VVP-I na mila ho, taaki wo Kaho ke 'agar VVP-I na milta' waale scenario ka ek reasonable stand-in ban sake.

2. Sirf Before-Vs-After Kaafi Kyun Nahi

PDF 1-3 mein humne H1 test dekha tha — sirf treated gaon ka 2021-vs-2025 NDBI compare karna. Ye ek natural pehla kadam hai, lekin akele adhoora hai. Kyun?

Kyunki 2021 se 2025 ke beech, VVP-I ke alawa bhi bahut kuch ho sakta hai jo NDBI ko badal sakta hai — general economic growth, koi doosri state/central scheme, migration patterns, ya sirf normal population growth ke saath organic construction. Agar hum sirf treated gaon ka before-after change dekhte hain, hum ye sab confounding factors bhi apne 'VVP-I effect' mein galti se mila dete hain.

EXAMPLE — Weight-Loss Program Ki Analogy

Socho tu ek naya diet plan follow karti hai 3 mahine tak, aur 3 kg weight kam hota hai. Kya poora 3 kg diet plan ki wajah se hai? Shayad us dauraan tu zyada active bhi thi, ya weather change hui, ya koi aur factor bhi tha jo weight kam karta. Sirf 'before-after' dekhne se ye sab alag nahi kiye ja sakte — tujhe kisi aise insaan se compare karna padega jo same waqt, same conditions mein, lekin bina diet plan ke tha, taaki 'sirf diet plan ka effect' isolate ho sake. Isi tarah, VVP-I ka 'sirf scheme ka effect' isolate karne ke liye, humein aise gaon chahiye jo same waqt, same region mein, lekin bina VVP-I ke rahe hon.

Yahi wajah hai ki BORDER OPTICS ka H1 test (sirf treated villages ka before-after) hamesha ek incomplete tasveer deta hai — isliye H4 test (control group ke saath) zaroori hai, poori tasveer ke liye.

3. Control Group — Ek Practical Stand-In Counterfactual Ke Liye

Agar hum true counterfactual (time-travel wala) nahi paa sakte, agla best cheez hai: ek control group — aise units (yahan, villages) ka ek group jo treated group jitna similar ho sake, lekin jinko treatment (VVP-I) nahi mila. Treated group (jinko VVP-I mila) aur control group (jinko nahi mila) ka comparison, ek reasonable, practical counterfactual estimate deta hai.

Medicine se hi ek familiar term udhaar lete hain: treatment group (jisko dawai/intervention di gayi) aur control group (jisko nahi di gayi, ya placebo di gayi). Clinical trials mein researchers patients ko *randomly* in do groups mein baantte hain, taaki dono groups shuru mein similar hon — koi systematic farak na ho jo baad mein result ko confuse kare.

3.1 BORDER OPTICS Ka Control Group — Asli Design

VVP-I mein villages ko 'randomly' nahi chuna gaya — sarkaar ne specific, administrative criteria (border proximity, jaisa PDF 5 mein detail se dekhenge) se villages chuni. Isliye BORDER OPTICS random assignment nahi kar sakta (real random experiment nahi hai) — iske bajaye, ek carefully matched control group banaya gaya hai.

```
""" Builds the non-VVP control group: matched on district (verified against the real
Nominatim boundary, not just the Overpass bbox), soft filter at 1.5x treated max
border-distance, capped at 3x treated count per district. Fixed two ways a "control" could
secretly be a treated village: (1) distance filter now uses the same geodesic method as
compute_border_distance.py instead of a distorted single UTM zone, (2) exclusion now
also catches same-village-different-script duplicates (coord proximity check) and official
VVP-I villages that never got geocoded, not just exact name matches. """
```

Ye docstring hai asli `select_control_villages.py` file se (fresh-read, abhi). Control group banane ka logic hai: same 14 districts se non-VVP villages lo (taaki region-level factors — jaise district ka overall economic trend, infrastructure — dono groups mein common rahein), border-se-distance mein roughly similar range rakho (treated villages ke max distance ka 1.5x tak), aur per-district count ko treated count ke 3x tak cap karo (taaki koi ek district poore control group ko dominate na kare). Final control group mein 721 villages hain, 251 treated villages ke against.

Is docstring ka doosra half — jo 'Fixed two ways...' se shuru hota hai — khud is PDF ke agle section ki poori kahani summarize karta hai. Ek research script ka docstring sirf 'ye kya karta hai' nahi batata — kayi baar wo 'ye pehle kya galat kar raha tha, aur ab kaise fix hua' bhi record karta hai, taaki koi bhi jo baad mein is code ko padhe (khud Sakshi bhi, mahino baad), turant samajh jaaye ki ye specific check kyun hai, aur usko hatana kitna risky hoga.

EXAMPLE — Classmate Comparison Ki Analogy

Agar tu ye pata karna chahe ki ek naya coaching class tere marks badhane mein kaam aaya, sabse acha comparison hoga tere aise classmates se jo same school, same class, same starting-level marks ke the, lekin usi coaching class mein nahi gaye — na ki kisi random doosre shehar ke bachche se, jiski poori situation hi alag ho. BORDER OPTICS ka control group bhi yahi kar raha hai — non-VVP villages jo treated villages ke same districts mein hain (same broad region, same state-level policies), taaki comparison fair aur meaningful ho.

4. Ek Genuine Galti Ki Kahani — Control Group 'Contamination'

Ab ek real, important history-story, jo isliye zaroori hai kyunki ye dikhati hai control group design mein galti hona kitna aasan hai, aur usko pakadne/fix karne ka process kitna zaroori hai.

Development Log Entries 23-25 mein record hai: ek early version ke control group list mein 23 aisi rows thi jo secretly treated villages hi thin — 20 aise villages jo treated villages ke exact coordinate-duplicates the (bas alag script/spelling mein likhe hue, jaise Hindi transliteration ka farak), aur 3 aur jo officially VVP-I priority list mein the lekin geocode nahi ho paaye the, isliye 'control' list mein galti se reh gaye.

EXAMPLE — Kyun Ye Itna Bada Problem Hai

Socho tu apne coaching-class experiment mein, 'non-coaching' comparison group mein galti se kuch aise students bhi daal deti hai jo actually coaching class mein the hi — bas unka naam thoda alag likha gaya tha attendance sheet mein. Ab jab tu 'coaching vs non-coaching' marks compare karegi, 'non-coaching' group ka average bhi coaching ke effect se thoda upar khinch jaayega — kyunki wo group asal mein *part-coaching, part-non-coaching* hai. Result: coaching ka real effect underestimate ho jaayega, ya kabhi-kabhi poora pattern hi galat direction mein dikh sakta hai.

BORDER OPTICS mein exact yahi hua — jab in 23 'secretly-treated' villages ko control group mein rakha gaya, full-year night-lights DiD result significant ($p = 0.0364$, coefficient $+0.1653$) aaya tha — matlab lag raha tha ki treated villages, control se zyada roshan hue. Ye result kabhi ek genuine, real effect nahi tha — ye sirf ek data-quality bug ka artifact tha: treated villages ka apna trend, galti se, control side par bhi count ho raha tha, ek 'fake gap' banate hue jo asal mein tha hi nahi.

4.1 Fix Ek Din Mein Nahi Hua — Teen Alag Steps Mein Hua

Ye history ek single 'bug mila, fix kiya' waaki chhoti kahani nahi — asli research mein galtiyan aksar *layers mein* pakdi jaati hain, ek fix dusre ko expose karta hai. Poori timeline:

EXAMPLE — Step 1 — Entry 23: Problem Discover Hua

Ek full audit mein pata chala ki control-group script distance calculate karne ke liye ek galat coordinate-system (UTM 44N) use kar raha tha, jabki poora study area (Ladakh se Arunachal Pradesh tak, ~20 degrees longitude) ke liye wo distorted hai. Isi audit mein, ek aur, bada discovery hua: coordinate-matching se dekha gaya ki 20 control villages, treated villages ke *exact same coordinates* par baithe the — bas alag naam spelling mein (jaise 'Sibia'/'Sebia'), ya alag script mein (jaise 'Gunji' — jiska ek alag-script alt-name tag bhi tha OSM mein, ek hi physical gaon ke liye). Plus

3 aise villages jo VVP-I ki official list mein the lekin kabhi geocode nahi ho paaye — isliye unka naam kabhi 'treated names' list mein aaya hi nahi, aur wo control list mein reh gaye.

EXAMPLE — Step 2 — Entry 24: Code Fix Hua

Do fixes likhe gaye: (1) distance ab geodesic method (pyproj Geod, WGS84) se calculate hoti hai — wahi method jo treated villages ke liye already use ho raha tha — na ki distorted UTM se. (2) Ek naya 50-meter coordinate-proximity check add hua: agar koi candidate control village kisi treated village ke 50m ke andar hai, use same settlement maana jaata hai, naam kuch bhi ho. Plus ek fuller official-name check, jo sirf geocoded treated names tak limited nahi, poori official VVP-I list (552 names) se compare karta hai.

EXAMPLE — Step 3 — Entry 25: Fix Ko Live Data Par Chalaya Gaya

Code fix likhna ek cheez hai, use asli data par chalana doosri. Control list poori tarah regenerate hui — 735 se 732 villages tak (sirf 3 kam, kyunki naye filters candidate-pool level par kaam karte hain, seedhi subtraction nahi). Dono contamination checks ab zero aaye (pehle 20 aur 3 the). Poora DiD dobara chalaya gaya: full-year lights coefficient $+0.1653$ ($p=0.0364$) se seedha $+0.05046$ ($p=0.30946$) par aa gaya — significant se null.

EXAMPLE — Step 4 — Entry 37: Ek Aur, Chhota Refinement

Baad mein, ek district-boundary-verification pass mein pata chala ki 2 districts (Tawang aur Pithoragarh) ke liye control-selection sirf ek rough bounding-box use kar raha tha, asli district boundary polygon nahi — kyunki Nominatim se un districts ka polygon pehle resolve nahi hua tha. Jab ye fix hua aur asli boundary polygon mila, 11 Tawang candidates aise nikle jo bbox ke andar the lekin asli district boundary ke *bahar* — unko hataya gaya. Control count 732 se 721 par aa gaya, aur full-year lights DiD final number ban gaya coefficient $+0.03460$, $p = 0.45334$ — wahi number jo is PDF mein Section 5.1 mein pehle dikh chuka hai.

Yahi hai 721 villages ka poora origin — ek 735-village list se shuru hoke, do alag contamination-fixing rounds (Entry 25: 735→732) aur ek boundary-verification round (Entry 37: 732→721) se guzarke. Har step mein number thoda change hua, lekin final conclusion — full-year lights DiD null hai — ek baar bhi nahi badla in dono fixes ke beech mein.

EXAMPLE — Interesting Twist — Ek Fix Ne Dusri Cheez Bhi Change Ki

Entry 25 ke fix ke turant baad (732-village list par), full-year lights baseline-balance check (2021 mein treated vs control already similar the ya nahi) *balanced* ($p=0.08976$) nikla tha — ek achhi sign. Lekin Entry 37 ke boundary-refinement ke baad (721-village list par), wahi check *phir se imbalanced* ho gaya ($p=0.03558$, jo is PDF ke Section 6 mein dikhega). DiD result khud ($p=0.45334$) nahi badla — lekin us result ki 'baseline foundation' kitni mazboot hai, ye do alag control-list versions mein alag-alag dikh raha hai. Yahi wajah hai research paper hamesha specific batata hai ki

kaunsa number kis control-list version se aaya — 'null hai' kehna kaafi nahi, *kis data par null hai* bhi utna hi zaroori hai.

Ye kahani is PDF ke liye ek perfect real-world lesson hai: DiD ka poora result control group ki quality par depend karta hai. Agar control group mein koi bhi treated-jaisi cheez mix ho jaaye, poora comparison bigad jaata hai — chahe statistics ka formula bilkul sahi kyun na ho.

5. Difference-in-Differences (DiD) — Do Baar Farak Nikaalna

Ab poore concept ka naam samajhte hain: Difference-in-Differences. Naam mein hi logic chhupa hai — ye sirf ek farak nahi, do farak ka farak nikaalta hai.

Pehla farak: treated group ka before-vs-after change (jaise Kaho ka 2021→2025 NDBI change). Doosra farak: control group ka before-vs-after change (usi waqt ke non-VVP villages ka change). DiD ka final number hai in dono farakon ka farak — treated ka change, minus control ka change. Ye final number batata hai: treated villages, control villages se *kitna zyada* (ya kam) badle — aur ye theory hai ki jo bhi 'common' factors dono groups par equally asar karte (general growth, regional trends), wo is subtraction mein cancel ho jaate hain, sirf VVP-I-specific effect bachta hai.

EXAMPLE — Do Dukanon Ki Analogy

Socho do same-size dukanein hain, ek hi bazaar mein. Ek dukaan ne naya advertisement lagaya (treatment), doosri ne nahi (control). Agar advertisement-wali dukaan ki sales 20% badhi, kya ye sab advertisement ki wajah se hai? Shayad poore bazaar mein hi sales badh rahi thi (festival season, jaise). Agar hum dekhen ki non-advertisement wali dukaan ki sales bhi 12% badhi (same bazaar, same season ka effect), toh advertisement ka 'extra' effect hai: $20\% - 12\% = 8\%$. Yahi DiD ka poora logic hai — 'total change' mein se 'jo bhi vaise hi hota' ko subtract karke, sirf 'treatment-specific extra change' nikaalna.

5.1 Asli Regression Mein Kaise Dikhta Hai

```
formula_fe = f"{outcome} ~ treatment + post + did_term + C(district)"
model_fe = smf.ols(formula_fe, data=valid).fit(cov_type="cluster", cov_kwds={"groups":
valid["district"]})
```

Ye asli code hai did_model.py se (fresh-read). treatment (0 ya 1, gaon treated hai ya control) aur post (0 ya 1, ye 'before' period hai ya 'after') dono alag-alag include hote hain, lekin sabse zaroori variable hai did_term — jo $\text{treatment} \times \text{post}$ hai. Ye interaction term hi asli 'DiD coefficient' deta hai — iska matlab hai *specifically* 'treated hone AND post-period hone' ka combined, extra effect — na sirf treated hone ka, na

sirf post-period hone ka, balki dono ek saath hone ka.

EXAMPLE — BORDER OPTICS Ke Asli, Current DiD Numbers

Full-year NDBI: did coefficient = +0.00668, 95% CI [-0.0057, +0.0191], $p = 0.29102$ — not significant (CI mein 0 shaamil hai, PDF 3 yaad kar). Summer NDBI: coefficient = -0.00275, CI [-0.0114, +0.0059], $p = 0.53098$ — bhi not significant. Full-year lights: coefficient = +0.03460, CI [-0.0558, +0.1250], $p = 0.45334$. Summer lights: coefficient = +0.01798, CI [-0.0583, +0.0943], $p = 0.64393$. Chaaron combinations mein CI zero ko cross karta hai, aur chaaron p-values 0.05 se bahut zyada hain — matlab treated villages, control villages se statistically alag tarah se nahi badle, kisi bhi outcome ya window mein. Ye H4 ka headline null result hai.

5.2 Ek Haath-Se-Chalane-Waali DiD Example

Formula se pehle, ek bahut chhota, hypothetical example (asli 251/721 nahi, sirf logic samajhne ke liye):

Group	NDBI 2021	NDBI 2025	Change
Treated	0.10	0.14	+0.04
Control	0.08	0.10	+0.02

Treated group ka change: +0.04. Control group ka change: +0.02. $DiD = +0.04 - +0.02 = +0.02$. Iska matlab: treated villages, control villages se 0.02 extra badhe — ye wo hissa hai jo sirf treatment se attribute ho sakta hai (agar parallel trends assumption sahi hai). Agar treated group ka change +0.04 tha aur control group ka bhi +0.04 hota (same), $DiD = 0$ hota — matlab poora +0.04 'general trend' tha, VVP-I ka koi extra contribution nahi.

EXAMPLE — Ek Zaroori Twist — Negative Control Change Bhi Possible Hai

BORDER OPTICS ke asli control group mein kayi outcomes mein control ka change treated se bhi zyada ho sakta hai, ya negative ho sakta hai — jaise agar control villages mein NDBI thoda ghata. Iska matlab treatment 'negative' hai, ye zaroori nahi — ho sakta hai control region mein hi koi unrelated cheez chal rahi ho (jaise ek local factor). DiD sirf ek relative comparison deta hai, absolute 'achha' ya 'bura' judgment nahi — jo bhi doosre group se extra/kam hua, wahi number nikaalta hai, uski wajah interpret karna alag kaam hai.

6. Parallel Trends — DiD Ki Sabse Zaroori (Aur Sabse Fragile) Assumption

DiD ka poora logic tab hi sahi hai jab ek key assumption sach ho: parallel trends — matlab, agar VVP-I na hota, treated aur control villages, dono ka trend (trajectory) *same direction, same rate* mein chalta rehta. DiD ye directly prove nahi kar sakta (kyunki hum counterfactual nahi dekh sakte — Section 1 yaad kar), lekin agar ye assumption galat ho, poora DiD result meaningless ho sakta hai.

EXAMPLE — Do Gaadiyon Ki Race Ki Analogy

Socho do gaadiyan ek highway par chal rahi hain, roughly same speed se. Agar ek gaadi ko turbo-boost (treatment) milta hai aur wo aage nikal jaati hai, hum keh sakte hain boost ne kaam kiya — *agar hum jaante hain ki bina boost ke bhi dono gaadiyan same speed maintain karti*. Lekin agar ek gaadi pehle se hi thodi tezz thi (already-diverging trend), uska aage nikalna boost ki wajah se ho sakta hai, ya sirf uski already-different trajectory ki wajah se — hum confidently nahi keh sakte. Parallel trends assumption yahi check karti hai: kya dono gaadiyan, boost milne se pehle, sach mein 'parallel', same-trajectory mein chal rahi thi.

6.1 BORDER OPTICS Ismein Kya Kar Sakta Hai — Aur Kya Nahi Kar Sakta

Ek 'proper' parallel-trends check ke liye multiple *pre-treatment* time-points chahiye (jaise 2018, 2019, 2020, 2021 — sab treatment se pehle), taaki dekha ja sake ki treated aur control villages us poore pre-period mein similar trend follow kar rahe the ya nahi. BORDER OPTICS ke control group ke paas ye multi-year pre-treatment data nahi hai — sirf ek single 'before' point (2021) hai.

Iske bajaye, BORDER OPTICS ek weaker, lekin still useful, stand-in check karta hai: baseline balance — 2021 mein hi, treatment se pehle, kya treated aur control villages already similar levels par the (Mann-Whitney U test, PDF 3 mein seekha). Agar dono groups 2021 mein already bahut alag hain, ye ek red flag hai ki wo shayad different trajectories par bhi honge.

EXAMPLE — Real, Honest Result — Ek Baseline Imbalance

Teen combinations mein baseline balanced hai (full-year NDBI $p=0.181$, summer NDBI $p=0.461$, summer lights $p=0.088$). Lekin ek combination mein — full-year night lights — treated aur control 2021 mein hi statistically alag the (treated mean 0.42821 vs control 0.43493, $p = 0.03558$). BORDER OPTICS ise chhupata nahi — khule taur par bolta hai ki full-year lights DiD ka null result (jo khud mein sahi hai) is ek combination ke liye utni mazboot foundation par nahi khada, jitni baaki teeno par. Ye exactly wo honest, transparent reporting hai jo achhi research ko define karta hai — apni khud ki assumption ki limitation ko chhupana nahi, khule taur par batana.

7. District Fixed Effects — Har District Ka Apna 'Baseline' Hata Dena

Formula mein ek aur piece tha: $C(\text{district})$ — ye district fixed effects add karta hai. Iska matlab: regression, har district ke liye apna alag, separate 'starting level' allow karta hai, aur sirf *within-district* change ko compare karta hai.

EXAMPLE — 14 Alag Classrooms Ki Analogy

Socho tu 14 alag classrooms ke students compare kar rahi hai, exam-score-improvement dekhne ke liye. Har classroom ka apna, alag average starting-score ho sakta hai (ek classroom mein pehle se hi tez students the, doosre mein nahi). Agar tu seedhe sabko mila ke compare kare, classroom-level farak, tere asli treatment-effect ke saath confuse ho sakta hai. District fixed effects bilkul yahi karte hain — har classroom (district) ko uski apni baseline ke against compare karte hain, taaki sirf treatment ka effect bache, districts ke beech ke pehle-se-maujood farak nahi.

BORDER OPTICS mein 14 districts hain (treated aur control dono usi 14 mein se), aur fixed effects ye ensure karte hain ki agar ek district (jaise Tawang) overall zyada ya kam developed ho, wo district-level farak DiD estimate ko bias na kare — sirf 'is district ke andar, treated villages control villages se kitna zyada/kam badle' hi count hota hai.

7.0 Asli Data Mein Districts Kitne Alag Dikhte Hain — Ek Fresh Example

Ye abstract nahi hai — BORDER OPTICS ke per-district breakdown (fresh-read, `border_optics_did_by_district_fullyear.csv` se) mein districts genuinely bahut alag-alag dikhte hain. Full-year night-lights change (2021→2025) do districts mein:

District Treated Δ Control Δ Gap Chamoli +1.18330 +0.11507 +1.06823 Kurung Kumey
-0.02213 +0.00704 -0.02916

Chamoli (Uttarakhand) mein treated villages ka lights change +1.18 hai — bahut bada, positive jump. Kurung Kumey (Arunachal Pradesh) mein treated villages ka change sirf -0.02 hai — halka sa girav. Agar hum bina fixed effects ke, sab districts ko ek saath mila ke compare karte, Chamoli jaisa ek extreme district poore average ko khinch sakta tha, aur asli, within-district treatment-effect chhup jaata. District fixed effects isi extreme district-to-district variation ko 'absorb' kar lete hain, taaki DiD sirf ye poochhe: *har district ke andar*, treated villages apne hi district ke control villages se kitna zyada/kam badle — Chamoli ka apna baseline alag, Kurung Kumey ka apna, aur dono fairly treated hote hain apne hi comparison ke against.

7.1 Clustering — Ek Chhota, Zaroori Technical Detail

`cov_type="cluster", cov_kwds={"groups": valid["district"]}` — ye standard errors ko district ke hisaab se cluster karta hai. Iska matlab: same district ke villages ko completely independent nahi maana jaata (wo ek doosre se related ho sakte hain — same weather, same local economy), balki district-level correlation ko account mein liya jaata hai, taaki confidence intervals aur p-values zyada honest (na zyada confident, na kam) hon. BORDER OPTICS khud khule taur par ye bhi disclose karta hai ki 14 clusters, standard 30-40+ ki recommendation se kam hai — isliye ek comparison

HC3 (no fixed effects) version bhi report hota hai, aur PDF 14 mein ek aur stress-test (wild cluster bootstrap) bhi dekhenge, jo specifically ye chhoti-cluster-count wali concern address karta hai.

7.2 Selection Bias — Ek Aur Zaroori Concept

Ek aakhri, bahut zaroori concept: selection bias. Ye tab hota hai jab treatment kisi non-random reason se assign hoti hai jo khud outcome se bhi related ho. VVP-I villages randomly nahi chuni gayin — unhe specifically *border proximity* ke aadhar par chuna gaya (jaisa PDF 5 mein dekhenge). Agar border-proximity khud NDBI-growth se related hai (kisi aur reason se, treatment ke alawa), toh treated aur control villages ka simple comparison bhi bias ho sakta hai.

EXAMPLE — Hospital Ki Classic Example

Ek famous example: agar tu dekhe ki jo log hospital jaate hain unki health, na-jaane-waalo se kam achhi hoti hai, kya iska matlab hospital jaana health kharab karta hai? Bilkul nahi — log hospital isliye jaate hain kyunki wo pehle se hi bimaar hote hain (selection into treatment khud, outcome-related reason se hoti hai). VVP-I mein bhi similar risk hai — agar border-ke-paas gaon already kisi aur wajah se (jaise strategic importance ki wajah se pehle se zyada infrastructure investment) different the, treated-vs-control comparison confound ho sakta hai. Yahi wajah hai ki H3 (border proximity test, PDF 13) aur H4 (control group DiD) dono, ek doosre ko cross-check karte hain — agar dono alag kahaniyan batayein, ye interesting hai; agar same kahani batayein, confidence badhti hai.

District fixed effects (Section 7) selection bias ko *partially* address karte hain (agar selection district-level factors se juda ho), lekin poori tarah nahi khatam karte — isliye BORDER OPTICS is limitation ko bhi khule taur par disclose karta hai, apne research paper mein.

8. Ek Aage Ki Jhalak — Placebo Tests

Agar parallel-trends assumption directly test nahi ho sakti (Section 6), ek indirect tareeka hai: placebo test — DiD ko ek aise time-period par chalana jahan hum *jaante hain* koi treatment nahi hua tha (jaise 2019 se 2021, VVP-I approval se pehle). Agar wahan bhi 'significant gap' dikhta hai, ye ek red flag hai ki humara design kisi aur cheez ko pakad raha hai, VVP-I ko nahi. BORDER OPTICS ne bilkul ye test kiya hai — hum ise PDF 15 mein detail se dekhenge, jab hum `pretreatment_placebo_test.py` kholenge.

9. Common Confusions — Seedhe Sawaal, Seedhe Jawaab

9.1 'Agar DiD null hai, toh kya matlab VVP-I scheme fail hui?'

Nahi — aur ye ek bahut zaroori distinction hai. 'Satellite se koi detectable change nahi dikha' aur 'scheme fail hui' same cheez nahi hain. BORDER OPTICS ka apna Discussion section (Section 5) ye bilkul saaf bolta hai: ye result 'VVP-I ka investment kaam nahi kiya' waala claim nahi karta — ye sirf itna kehta hai ki 2021-2025 ke is specific window mein, is specific measurement approach se (satellite NDBI + night lights), koi statistically detectable ground-level physical change nahi mila, control group ke against.

EXAMPLE — Thermometer Ki Analogy

Agar tu ek thermometer se kisi cheez ka temperature measure kare aur wo 'no change' dikhaye, iska matlab ye nahi ki us cheez mein *kuch bhi* nahi badla — ho sakta hai koi cheez badli ho jo thermometer measure hi nahi karta (jaise, texture, ya smell). Waise hi, VVP-I ke asar ho sakte hain jo satellite NDBI/lights measure nahi kar sakti — jaise institutional changes, social outcomes, ya itna chhota physical change jo satellite ki resolution se neeche ho. BORDER OPTICS ka null result ek honest, specific claim hai: '*satellite audit se koi signal nahi mila*' — na ki ek overreaching claim ki 'kuch bhi nahi hua.'

Jo baat is null result ko genuinely important banati hai, wo hai context: VVP-I ka kabhi bhi koi official impact assessment nahi hua (Parliament record khud ye maanta hai). Toh ek multi-method satellite audit jo 'no detectable signal' batata hai, wo bhi ek valuable answer hai — absence of evidence bhi useful hai jab kahin aur koi evidence hi nahi hai.

9.2 'Control group treated villages jaisa 100% same hona chahiye?'

Nahi, aur asal mein ye *possible hi nahi hai* jab tak real random experiment na ho. BORDER OPTICS ka control group 'similar hona chahiye kuch specific, zaroori dimensions par' — same districts, roughly similar border-distance range — lekin har cheez match nahi ho sakti, kyunki villages randomly assign nahi hue the (Section 3 yaad kar). Yahi wajah hai baseline balance check, district fixed effects, aur Section 7.2 ka selection-bias discussion zaroori hain — *because* match perfect nahi hai, in sab checks se pata chalta hai match kitna acha hai, aur kahan weak ho sakta hai.

9.3 'Pehle ek significant result mila tha ($p=0.0364$) — usko turant report kyun nahi kar diya?'

Ye is poori PDF ka shayad sabse zaroori lesson hai. BORDER OPTICS mein do baar aisa hua ki ek 'significant' result mila, aur dono baar wo asal mein ek data-bug ka artifact nikla — ek toh yahi control-group-contamination waala (Section 4), doosra summer-window ka wo incomplete-Sikkim-data waala case jo PDF 1 mein dekha tha. Dono baar, jaise hi result significant dikha, usko turant paper mein finalize nahi kiya gaya — balki *usi hi scrutiny se guzara gaya* jo har doosre result ko milti hai: kya sample complete hai? Kya control group clean hai? Kya leave-one-out mein result stable rehta hai? Dono baar, us extra scrutiny ne hi bug pakda.

EXAMPLE — Ek Achhi Research Habit

Socho tu apne exam mein ek surprisingly high score paati hai. Ek achha student sirf khush hoke aage nahi badhta — wo check karta hai ki kahin answer-key mein koi mistake toh nahi thi, kahin koi question skip toh nahi hua jo count nahi hua. Achhi research mein bhi yahi habit honi chahiye: ek naya, 'surprising' significant result *zyada* scrutiny deserve karta hai, kam nahi — kyunki wo exactly wo jagah hai jahan ek bug apne aap ko 'discovery' ki tarah disguise kar sakta hai.

9.4 'H1 aur H4 dono null aaye — kya ye same cheez do baar test kar rahe hain?'

Nahi — dono alag roles play karte hain, aur ye zaroori hai samajhna. H1 (PDF 1-3 mein dekha, treated villages ka apna before-after change) is study ka primary estimand hai — main sawaal jispar paper ka headline conclusion tika hai. H4 (ye DiD test, control group ke saath) ek pre-specified robustness check hai — H1 ke result ko stress-test karne ke liye banaya gaya, na ki usse replace karne ke liye. Agar H1 aur H4 alag baatein batate — jaise H1 mein increase dikhta lekin H4 mein control group bhi utna hi badhta — toh H4 humein batata ki H1 ka result shayad 'general trend' tha, sirf VVP-I nahi. Dono null aana ek *consistent, mutually-reinforcing* tasveer deta hai — do alag tareekon se, ek hi conclusion tak pahunchna, jo kisi ek test se zyada confidence deta hai.

Is PDF Ka Recap — Ab Tak Kya Seekha

- Counterfactual ('agar treatment na hota toh kya hota') kabhi directly observe nahi hota — yahi causal inference ka fundamental problem hai.
- Sirf before-vs-after (H1) confounding factors (general growth, doosri schemes) se contaminate ho sakta hai — isliye control group zaroori hai.
- Control group ek practical counterfactual stand-in hai — BORDER OPTICS mein 721 non-VVP villages, same 14 districts se, carefully matched.
- Ek real history: 23 contaminated (secretly-treated) control villages ne ek fake-significant result diya tha; fix hone ke baad wo result null ho gaya — control-group quality DiD ka poora foundation hai.
- DiD = treated ka before-after change, MINUS control ka before-after change — 'common factors' cancel ho jaate hain, sirf treatment-specific effect bachta hai.
- BORDER OPTICS ke asli DiD numbers (NDBI aur lights, dono windows) sab not-significant hain, CI mein hamesha 0 shaamil hai.
- Parallel trends assumption (treated/control same trajectory follow karte agar treatment na hota) DiD ki sabse zaroori, sabse fragile assumption hai — poori tarah test nahi ho sakti, sirf baseline balance se partially check hoti hai.
- District fixed effects district-level baseline farak hata dete hain; clustering standard errors ko district-level correlation ke liye honest banata hai.
- Selection bias ek zaroori risk hai jab treatment randomly assign nahi hoti — district fixed effects ise sirf partially address karte hain.
- Null result ka matlab 'scheme fail hui' nahi — ye ek specific, honest claim hai: is method se koi detectable satellite signal nahi mila. H1 aur H4 dono null aana, ek hi conclusion ko do alag tareekon se confirm karta hai.

Aage kya: PDF 5 mein hum Big Picture dekhenge — VVP-I policy ka poora context, Parliament ka accountability gap, securitization theory, aur BORDER OPTICS ka poora research design (H1/H3/H4 kya test karte hain aur kyun, ek saath jode hue). Uske baad, PDF 6 se asli code files khulna shuru honge, geocoding se — aur jahan bhi koi statistical ya causal-inference concept milegi, jo PDF 3 aur 4 mein seekha, wahi turant kaam aayega.

Jab ready ho agle PDF ke liye, bas bol dena.